



Guangdong Technion

Israel Institute of Technology

广东以色列理工学院

035033-Intro to Sensor-Integrated Systems

Lecture VI: Sensing Elements

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Mechanical Engineering and Robotics

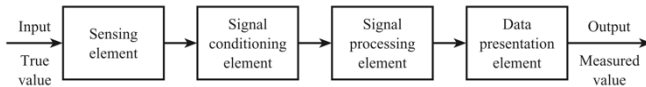
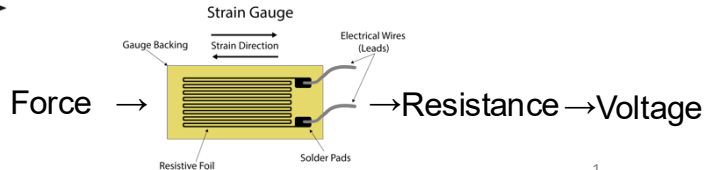
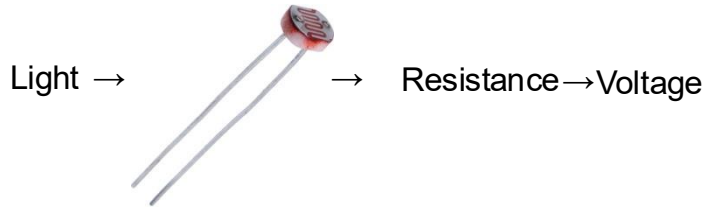
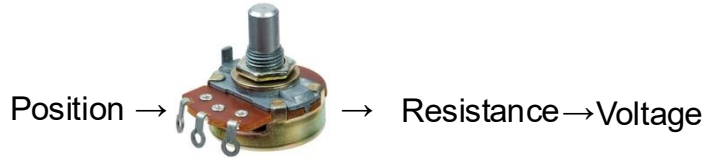
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Sensing element

- ❑ **Definition:** Sensing element is in contact with the process and gives an output which depends in some way on the variable to be measured
- ❑ The sensing element is the first element in the measurement system;
- ❑ Sensing element draws energy from the process or system being measured

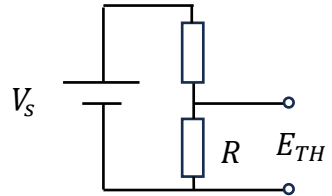
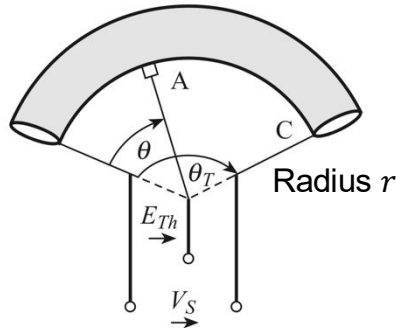


Resistive sensing elements - Potentiometers

Resistance $R = \frac{l}{\sigma A}$, where l is the length, A is cross section area, σ is conductivity, $\sigma = \frac{1}{\mu}$, μ is resistivity

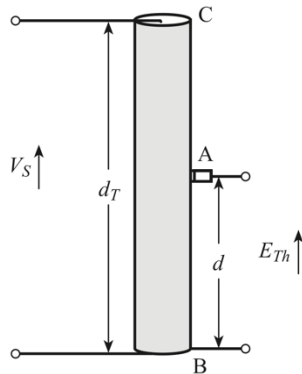
Rotational: $E_{TH} = \frac{R}{R_T} V_S = \frac{\frac{\theta r}{\sigma A}}{\frac{\theta_T r}{\sigma A}} V_S = \frac{\theta}{\theta_T} V_S$

$E_{TH} = f(\theta)$

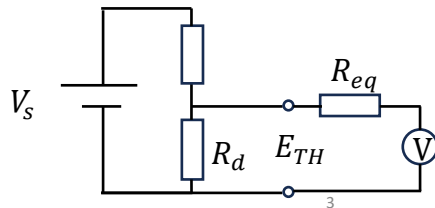
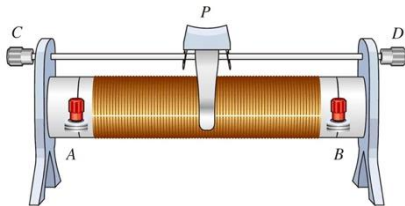


Resistive sensing elements - Potentiometers

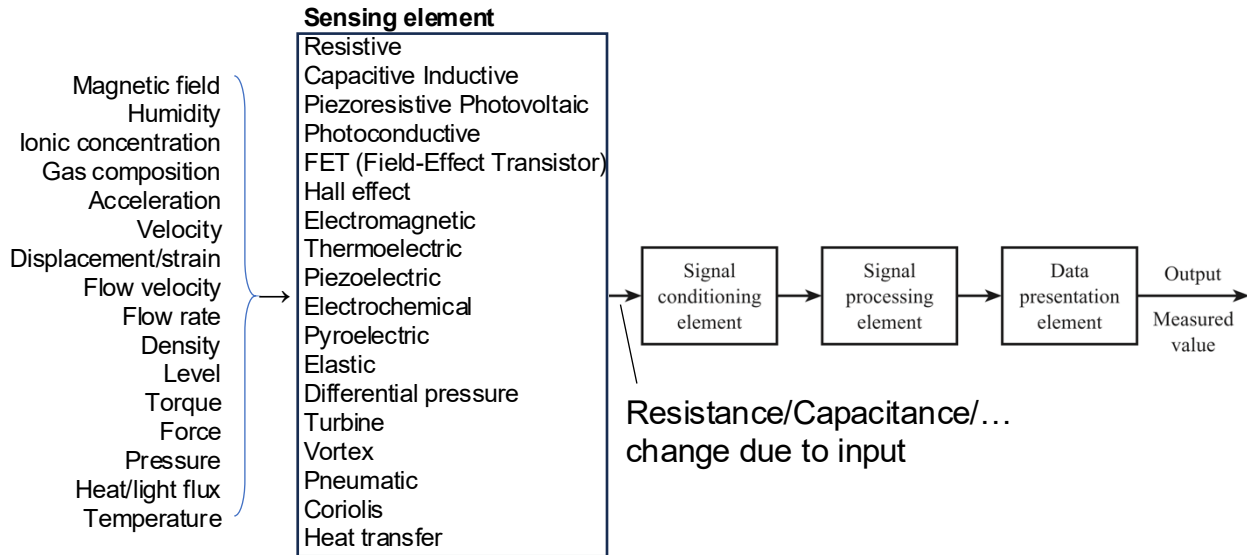
- ❑ Linear $E_{TH} = \frac{R_d}{R_T} V_S = \frac{\frac{d}{\sigma A}}{\frac{d_T}{\sigma A}} V_S = \frac{d}{d_T} V_S$
- ❑ Maximum travel d_T, θ_T : sensing range requirement
- ❑ Supply voltage V_S : output voltage requirement
- ❑ Power rating $W = \frac{V_S^2}{R_T}$ (to reduce power consumption, R_T should be larger)
- ❑ Resistance R_d : balance power consumption and load effect requirement



(a) Linear (rectilinear) displacement

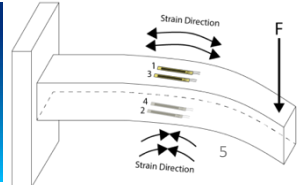
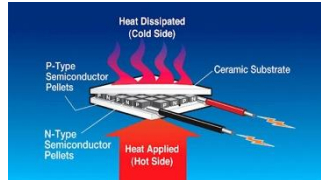
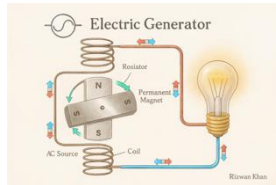
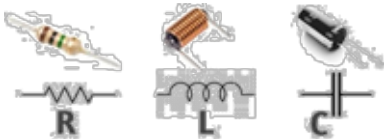


Overall Schematic



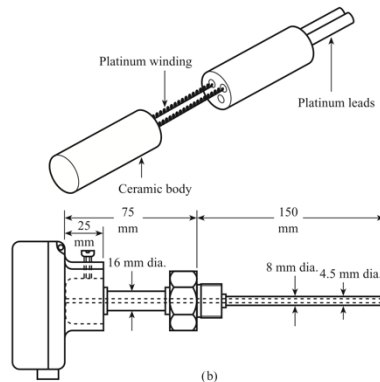
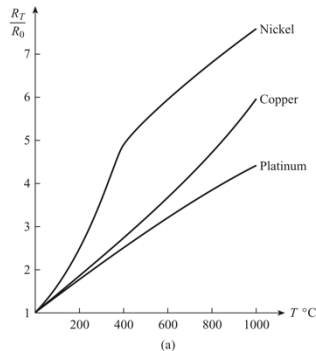
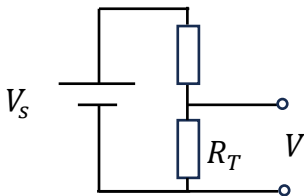
Different types of sensing element

- ❑ **Passive devices**, e.g., resistive, capacitive and inductive elements require an external power supply in order to give a voltage or current output signal;
- ❑ **Active devices**, e.g., electromagnetic and thermoelectric elements, need no external power supply
- ❑ Sensors with a **mechanical output** are commonly used as the primary sensing element in measurement systems for mechanical variables such as force or flow rate .
- ❑ In order to obtain an electrical signal, this primary element is followed by a **secondary sensing element** with an electrical output signal.
- ❑ **Examples1** : A resistive strain gauge sensing the strain in an elastic cantilever in a force measurement system (Force $\rightarrow$ Mechanical deflection $\rightarrow$ Resistance change)
- ❑ **Examples2** : An electromagnetic tachogenerator (测速发电机) sensing the angular velocity of a turbine in a flow measurement system.



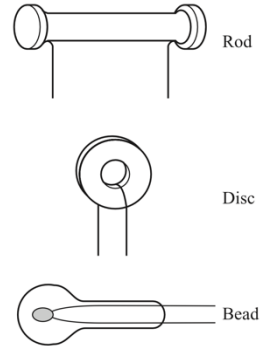
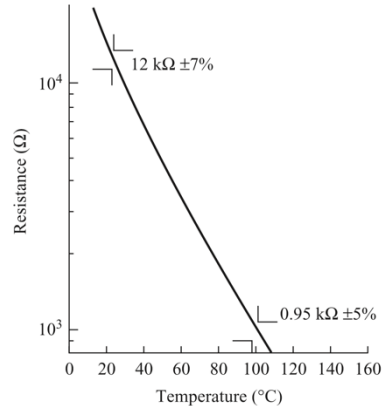
Resistive metal sensors for temperature measurement

- ❑ The resistance will change as the temperature changes
- ❑ Polynomial fitting: $R_T = R_0(1 + \alpha T + \beta T^2 + \gamma T^3 + \dots)$
- ❑ $R_T = f(T)$
- ❑ Read the voltage output, and measure temperature
- ❑ Nonlinearity is usually small $R_T = R_0(1 + \alpha T)$
- ❑ $V = f(R) = f(T)$



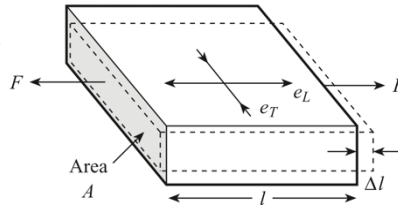
Different types of thermo-resistive sensors

- ❑ Semiconductor: The resistance at temperature $T^{\circ}K$: $R_T = K e^{\frac{\beta}{T}}$, K and β are constants for the thermistor
- ❑ Thick-film polymers: resistance change due to temperature

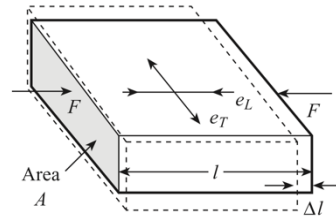


Resistive strain gauges

- ❑ Stress is defined by force/area, *i.e.*, $+\frac{F}{A}$, the positive sign: tensile stress tends to increase the length of the body (compressive stress is $-\frac{F}{A}$)
- ❑ Applied stress produce a strain in the body
- ❑ The tensile strain is $e = \frac{\text{change in length}}{\text{original unstressed length}} = +\frac{\Delta l}{l}$ (Right: compressive strain $-\frac{\Delta l}{l}$)
- ❑ Longitudinal tensile strain e_L , transverse tensile strain e_T ,
- ❑ Poisson's ratio: $\nu = -\frac{e_T}{e_L}$ (typically 0.25 ~ 0.4).
- ❑ Young's modulus $E = \frac{\text{stress}}{\text{strain}} = \frac{\text{stress}}{e}$
- ❑ A **strain gauge** is a metal element whose resistance changes when under strain.



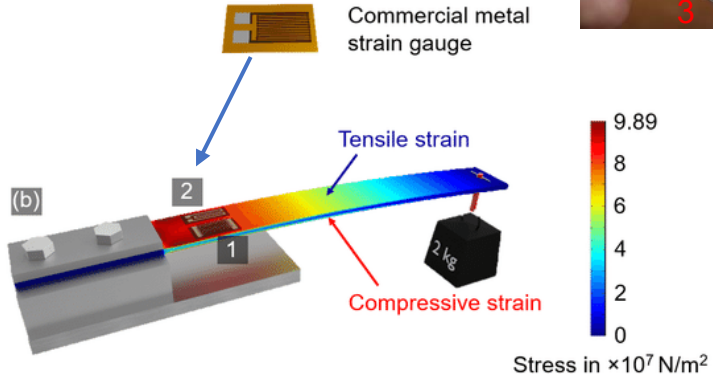
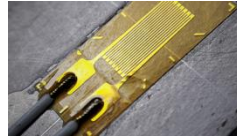
tensile stress



compression stress

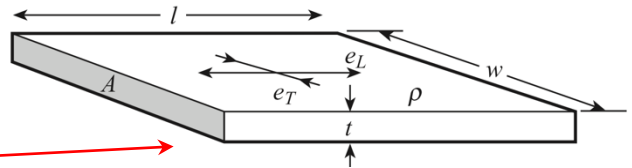
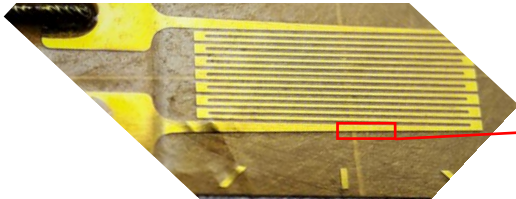
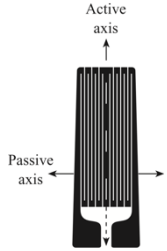
Example of strain gauge

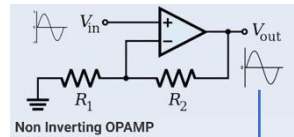
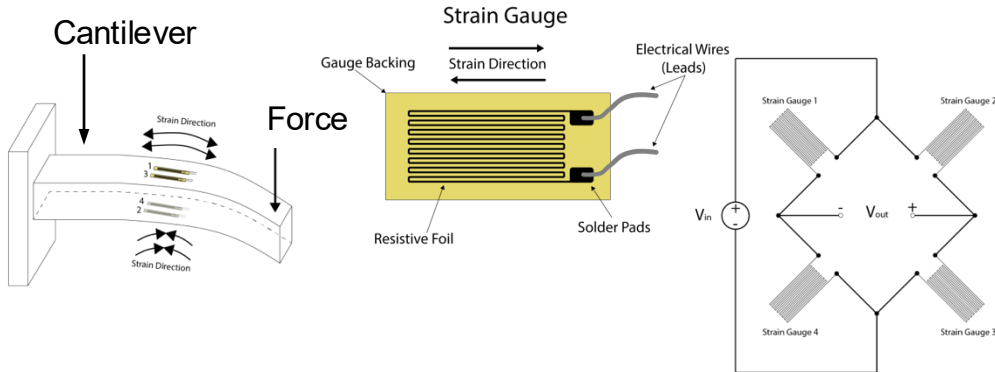
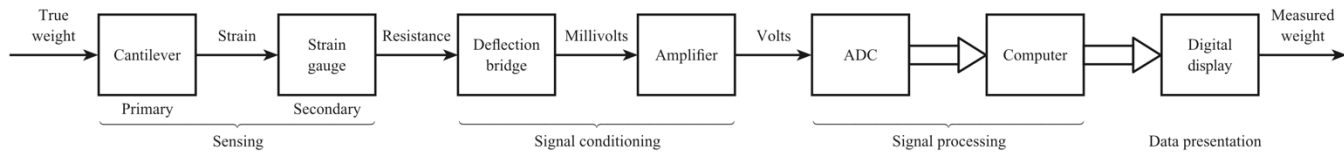
- ❑ The most common strain gauges are of the bonded type, where the gauge consists of metal foil, cut into a grid structure by a photoetching process, and mounted on a resin film base



Resistive strain gauges

- ❑ Resistance $R = \frac{\rho l}{A}$, where l is length, ρ is resistivity, A is cross section area
- ❑ $\Delta R = \frac{\rho}{A} \Delta l + \frac{l}{A} \Delta \rho - \frac{\rho l}{A^2} \Delta A \Rightarrow \frac{\Delta R}{R} = \frac{\Delta l}{l} + \frac{\Delta \rho}{\rho} - \frac{\Delta A}{A}$
- ❑ $A = wt \Rightarrow \frac{\Delta A}{A} = \frac{w \Delta t + \Delta w t}{wt} = \frac{\Delta t}{t} + \frac{\Delta w}{w} = -\nu e_L - \nu e_L = -2\nu e_L$
- ❑ $\frac{\Delta R}{R} = (1 + 2\nu) e_L + \frac{\Delta \rho}{\rho}$
- ❑ Define gauge factor $G = \frac{\frac{\Delta R}{R}}{e} = 1 + 2\nu + \frac{1}{e} \frac{\Delta \rho}{\rho} \approx 2$, (how much resistance change in ratio due to stress)
- ❑ $\frac{1}{e} \frac{\Delta \rho}{\rho}$ representing strain-induced changes in resistivity (piezoresistive effect) is small (for metal, it is around 0.4)



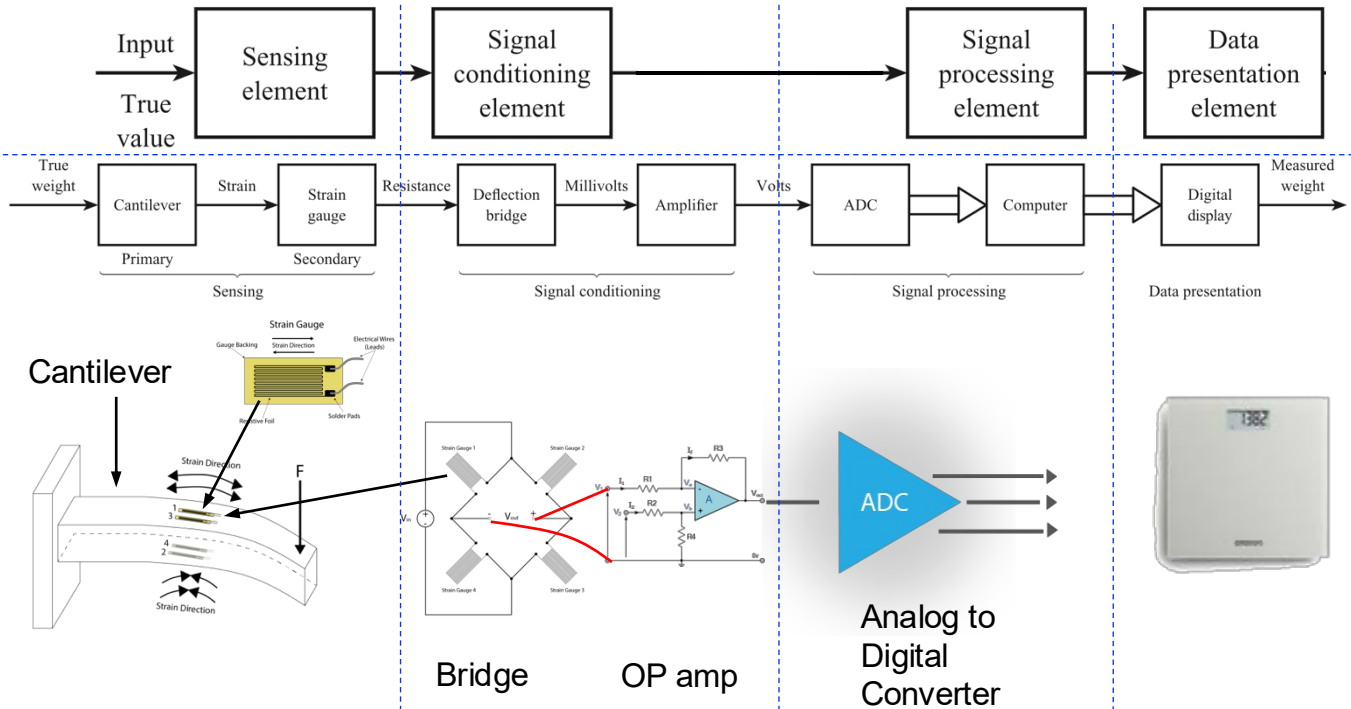


ADC



Full bridge

$$\begin{aligned}
 V_{out} &= V_+ - V_- = \frac{R - \Delta R}{R + \Delta R + R - \Delta R} - \frac{R + \Delta R}{R + \Delta R + R - \Delta R} = \frac{R - \Delta R - R - \Delta R}{2R} \\
 &= \frac{-2\Delta R}{2R} = -\frac{\Delta R}{R}
 \end{aligned}$$



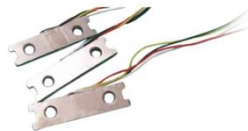
Final project

- ❑ Make a load sensor (scale) from scratch
- ❑ Conceive, debug, and finally make it work
- ❑ Compare with standard weight and do error analysis
- ❑ 6 questions and each cover a lecture in this semester

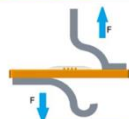


—使用场景展示—

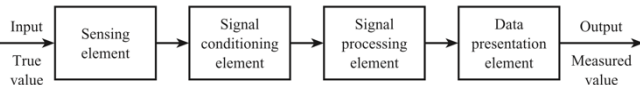
USE SCENARIO PRESENTATION



应变式称重测力传感器，利用惠斯通电桥原理把外部的压力通过传感器的弹性体变形使之贴上面的电阻式应变片发生阻值变化，在激励电压的作用下，输出与被测物成正比的模拟电信号。
传感器有三线制（半桥）、四线制（全桥）、五线制（带屏蔽线）
半桥传感器必须要偶数个组合使用
全桥传感器可以单个使用，也可以任意数量组合使用



固定端和受力端可以交换，请根据受力端位置固定和受力点同在中点。



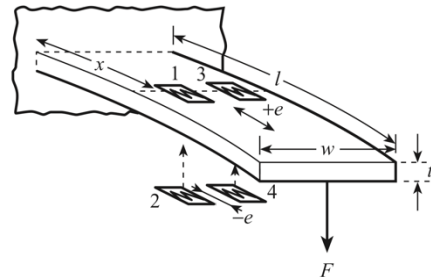
Four strain gauges are bonded onto a cantilever as shown in Figure 8.20(a). Given that the gauges are placed halfway along the cantilever and the cantilever is subject to a downward force of 0.5 N, use the data given below to calculate the resistance of each strain gauge:

Cantilever data

Length $l = 25 \text{ cm}$
Width $w = 6 \text{ cm}$
Thickness $t = 3 \text{ mm}$
Young's modulus $E = 70 \times 10^9 \text{ Pa}$

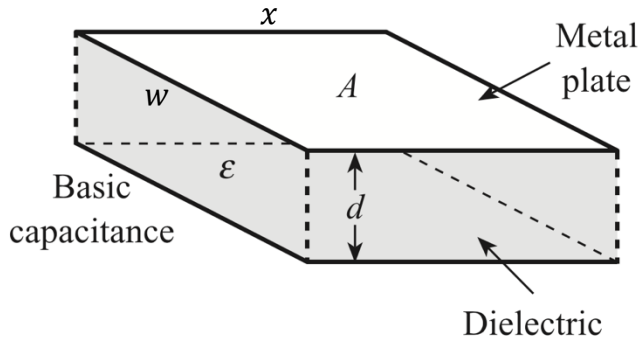
Strain gauge data

Gauge factor $G = 2.1$
Unstrained resistance $R_0 = 120 \Omega$

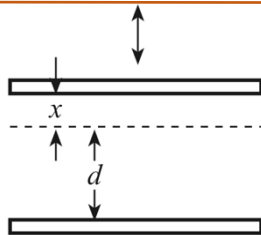


Capacitive sensing elements

- Capacitance $C = \frac{\epsilon_0 \epsilon A}{d}$, where ϵ_0 is the permittivity of free space (vacuum) of magnitude 8.85 pFm^{-1} , ϵ is relative permittivity of the insulating material, $A = x * w$ is the area of overlap of the plates, and d is their separation

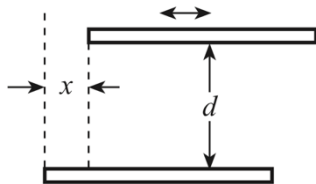


Capacitive sensing elements



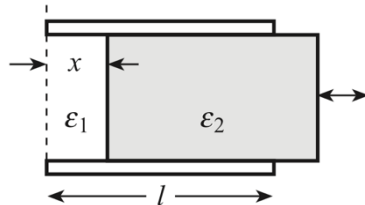
Variable separation

$$C = \frac{\epsilon_0 \epsilon A}{d+x}$$



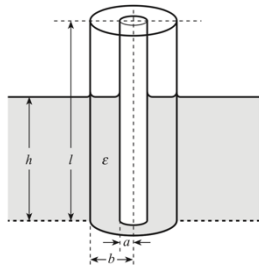
Variable area

$$C = \frac{\epsilon_0 \epsilon (A-wx)}{d}$$



Variable dielectric

$$\begin{aligned} C &= \frac{\epsilon_0 \epsilon_1 wx}{d} + \frac{\epsilon_0 \epsilon_2 (A - wx)}{d} \\ &= \frac{\epsilon_0}{d} (\epsilon_1 wx + \epsilon_2 (A - wx)) \\ &= \frac{\epsilon_0 w}{d} (\epsilon_2 l - (\epsilon_2 - \epsilon_1)x) \end{aligned}$$



Capacitive level sensor

$$C_h = \frac{2\pi\epsilon_0\epsilon h}{\log_e \frac{b}{a}} + \frac{2\pi\epsilon_0(l-h)}{\log_e \frac{b}{a}}$$

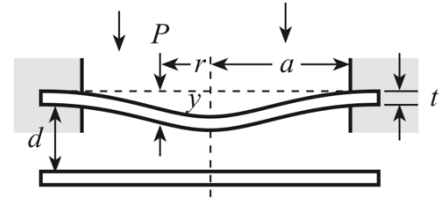
Cylindrical Capacitor



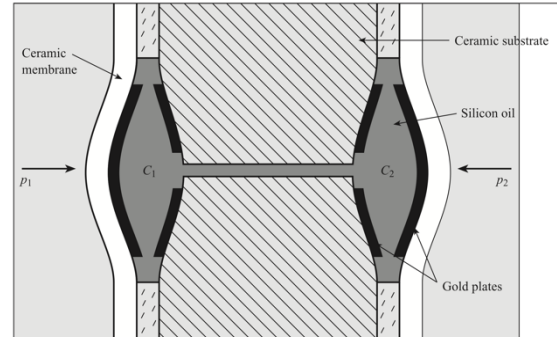
$$C = \frac{(2\pi\epsilon_0 \times L)}{(\ln \frac{b}{a})}$$

Typical configuration

- ❑ The dielectric constant ϵ for air is only 1, changes in capacitance ΔC is very small.
- ❑ To increase ΔC , the thickness t of the flexible plate/diaphragm can be reduced; this gives greater deformation y for a given pressure P but reduces mechanical strength.
- ❑ These problems can be largely overcome by using liquid-filled ceramic membrane sensors
- ❑ $\epsilon_{\text{silicon oil}} \gg \epsilon_{\text{air}} = 1$
- ❑ Dpressure $p_1 - p_2$ causes the left-hand membrane to move to the right, causing C_1 to increase and C_2 to decrease

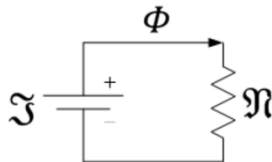
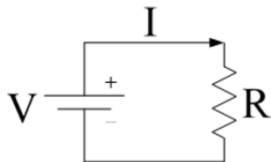
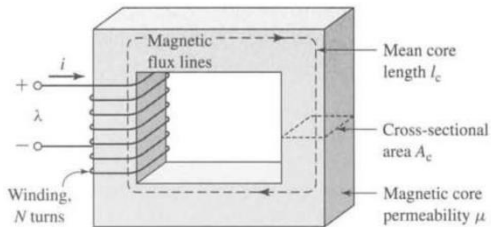
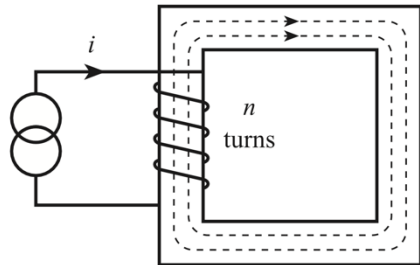


Capacitive pressure sensor



Inductive sensing elements

- ❑ Magnetomotive force m.m.f. $\mathcal{F} = Ni$
- ❑ Flux $\phi = \frac{\mathcal{F}}{\mathcal{R}} = \frac{Ni}{\mathcal{R}}$
- ❑ Reluctance $\mathcal{R} = \frac{l}{\mu A}$, where l is the length of core, μ : permeability, and A is cross section area
- ❑ Inductance $L = \frac{N^2}{\mathcal{R}}$



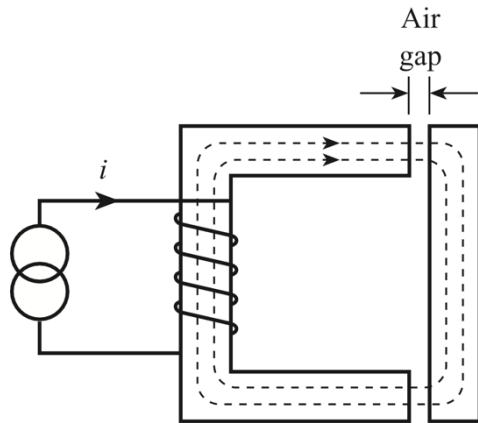
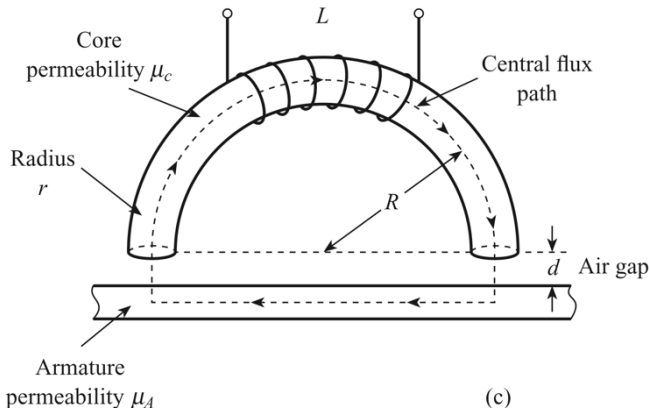
Inductive sensing elements

□ $\mathcal{R}_{\text{tot}} = \mathcal{R}_{\text{core}} + \mathcal{R}_{\text{gap}} + \mathcal{R}_{\text{amature}}$

□ $\mathcal{R}_{\text{tot}} = \frac{\pi R}{\mu_0 \mu_C \pi r^2} + \frac{2d}{\mu_0 \pi r^2} + \frac{2R}{\mu_0 \mu_A 2\pi r t} = \frac{R}{\mu_0 \mu_C r^2} + \frac{2d}{\mu_0 \pi r^2} + \frac{R}{\mu_0 \mu_A r t} = \mathcal{R}_0 + kd$

□ Where $\mathcal{R}_0 = \frac{R}{\mu_0 r} \left(\frac{1}{\mu_C r} + \frac{1}{\mu_A t} \right)$, $k = \frac{2}{\mu_0 \pi r^2}$

□ $\phi = f(\mathcal{R}) = f(d)$: inductive displacement sensor



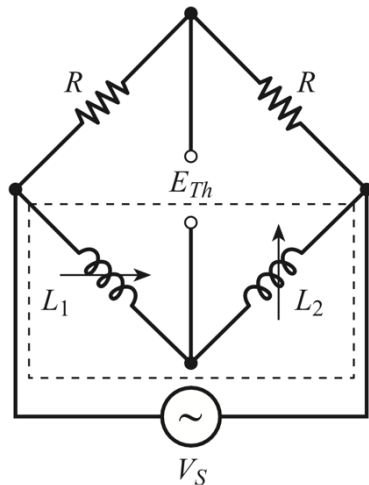
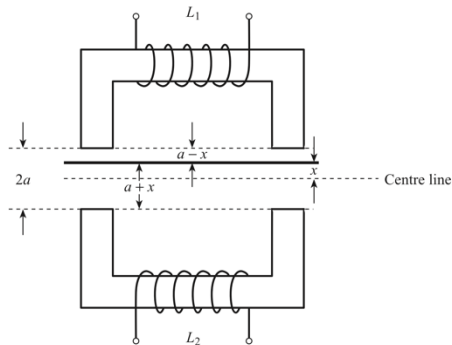
Bridge enable linearity

□ Inductance $L = \frac{N^2}{\mathcal{R}}$

□ $L = \frac{N^2}{\mathcal{R}_0 + kd} = \frac{L_0}{1 + \alpha d}$, where $\alpha = \frac{k}{\mathcal{R}_0}$

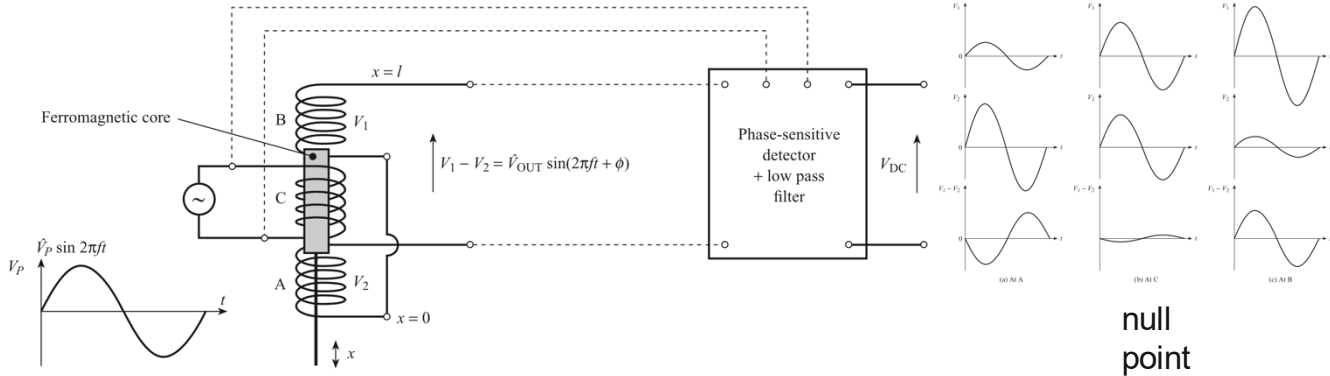
□ $L_1 = \frac{L_0}{1 + \alpha(a-x)}$, $L_2 = \frac{L_0}{1 + \alpha(a+x)}$

□ $E_{Th} = V_S \left(\frac{L_1}{L_1 + L_2} - \frac{1}{2} \right) = \frac{V_S \alpha}{2(1 + \alpha a)} x$



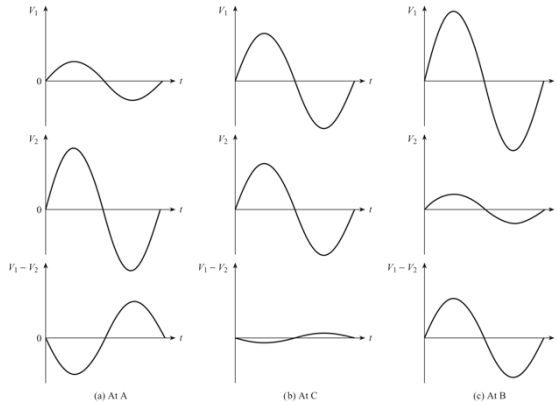
Linear Variable Differential Transformer (LVDT)

□ $V_{out} = V_1 - V_2 = f(x)$



Linear Variable Differential Transformer (LVDT)

- ❑ $V_{out} = V_1 - V_2 = f(x)$
- ❑ Amplitude and phase $\Rightarrow$ Amplitude and direction of D.C. output

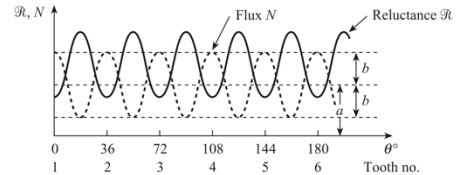
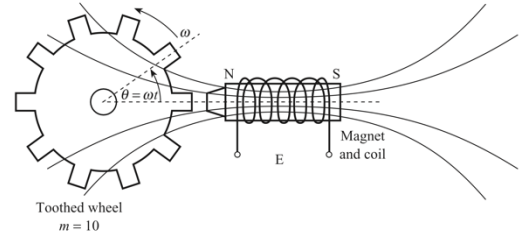


Electromagnetic sensing elements

$$\square \phi = \frac{m.m.f}{\mathcal{R}_{tot}}$$

$$\square \phi(t) = a + b \sin m\omega t$$

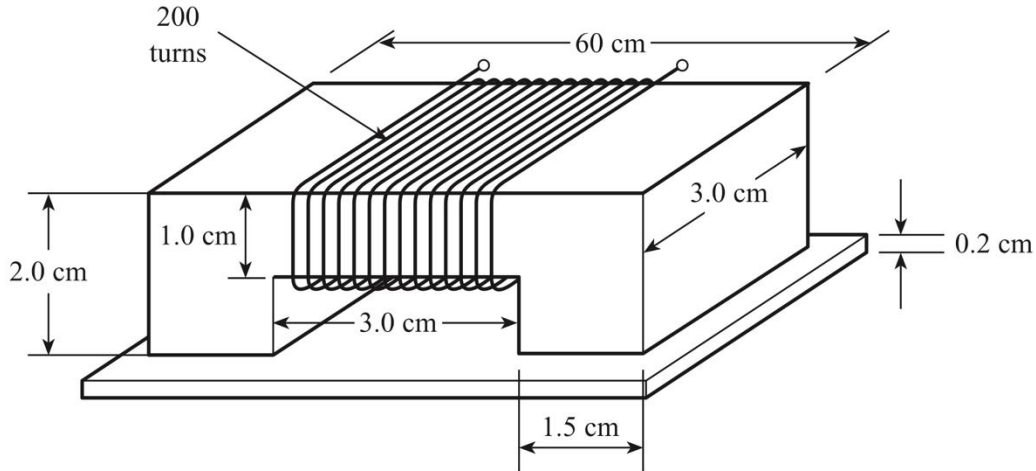
$$\square EMF = -N \frac{d\phi}{dt} = -Nm\omega b \cos m\omega t$$



By taking a central flux path, estimate the inductance of the sensor shown in Figure Prob. 5

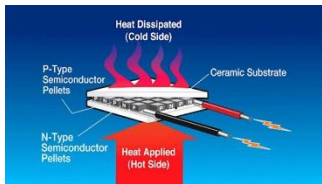
- (a) for zero air gap;
- (b) for a 2 mm air gap.

Assume the relative permeability of core and armature is 10^4 and that of air is unity.

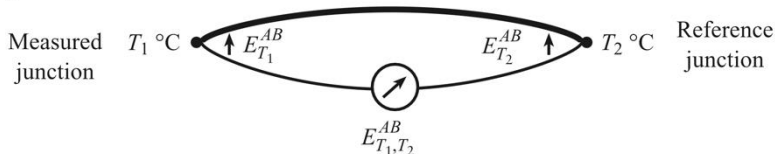


Thermoelectric sensing elements

- ❑ Thermoelectric or thermocouple sensing elements are commonly used for measuring temperature.
- ❑ $E_T^{AB} = a_1 T + a_2 T^2 + a_3 T^3 + \dots$
- ❑ $E_{T_1, T_2}^{AB} = a_1 (T_1 - T_2) + a_2 (T_1^2 - T_2^2) + a_3 (T_1^3 - T_2^3) + \dots$



Thermocouple circuit



An iron v. constantan thermocouple is to be used to measure temperatures between 0 and 300 °C. The e.m.f. values are as given in Table 8.2.

- (a) Find the non-linearity at 100 °C and 200 °C as a percentage of full scale.
- (b) Between 100 °C and 300 °C the thermocouple e.m.f. is given by $E_{T,0} = a_1T + a_2T^2$. Calculate a_1 and a_2 .
- (c) The e.m.f. is 12 500 μV relative to a reference junction of 20 °C and the corresponding reference junction circuit voltage is 1000 μV. Use the result of (b) to estimate the measured junction temperature.

Table 8.2 Thermocouple data and characteristics.

Type	Temperature ranges (°C)	e.m.f. values (μV) ^a	Tolerances	Extension leads	Characteristics
Iron v. constantan Type J	−20 to +700	$E_{100,0} = 5\,269$ $E_{200,0} = 10\,779$ $E_{300,0} = 16\,327$ $E_{500,0} = 27\,393$	Class 1 −40 °C to +375 °C ±1.5 °C	As for thermocouple	Reducing atmospheres have little effect. Should be protected from moisture, oxygen and sulphur-bearing gases

Elastic sensing elements

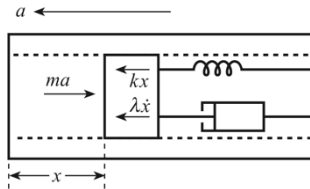
$$\square m\ddot{x} + \lambda\dot{x} + kx = ma \Rightarrow (ms^2 + \lambda s + k)x = ma \Rightarrow \frac{x}{a} = \frac{m}{ms^2 + \lambda s + k} = \frac{1}{s^2 + \frac{\lambda}{m}s + \frac{k}{m}} = \frac{\frac{1}{\omega_n^2}}{\frac{1}{\omega_n^2}s^2 + \frac{2\zeta}{\omega_n}s + 1}$$

$$\square \text{ Since the damping is small, } \frac{x}{a} = \frac{\frac{1}{\omega_n^2}}{\frac{1}{\omega_n^2}s^2 + 1}$$

$$\square \text{ Convert to frequency domain by defining } s = j\omega, \text{ then } \frac{x}{a} = \frac{\frac{1}{\omega_n^2}}{1 - \frac{\omega^2}{\omega_n^2}}$$

$\square$ The resonance freq. has to set to be much larger than the excitation freq. $\omega_n \gg \omega$

$\square$ Then $a = \omega_n^2 x = \frac{K}{m}x$ (what does this mean: we can obtain acceleration from the deflection amplitude.)



$$m\ddot{x} + \lambda\dot{x} + kx = ma$$

$$\frac{\Delta \bar{x}}{\Delta \bar{a}}(s) = \frac{\frac{1}{\omega_n^2}}{\frac{1}{\omega_n^2}s^2 + \frac{2\xi}{\omega_n}s + 1}$$

$$\omega_n = \sqrt{\frac{k}{m}}, \xi = \frac{\lambda}{2\sqrt{km}}, K = \frac{1}{\omega_n^2}$$

(a)



An accelerometer consisting of an elastic element and a potentiometric displacement sensor has to meet the following specification:

Input range = 0 to $5g$

Damping ratio = 0.8

Output range = 0 to 10 V

Maximum non-linearity = 2% of f.s.d.

Natural frequency = 10 Hz

Seismic mass = 0.005 kg

- (a) Calculate the required spring stiffness and damping constant.
- (b) What should the input displacement range of the potentiometer be? ($g = 9.81 \text{ m s}^{-2}$)
- (c) If the accelerometer is to be used with a recorder of $10 \text{ k}\Omega$ resistance, what is the maximum allowable potentiometer resistance?

Hall effect sensing element

- ❑ A rectangular slab of metal or semiconductor material and a corresponding xyz coordinate system. A current I flows through the slab in the x-direction, producing a corresponding current density J .
- ❑ A magnetic field of flux density B is applied to the slab in the z-direction. In a metal there are electron (n) charge carriers only; in a semiconductor there may be both electron and hole (p) charge carriers.
- ❑
$$E = -\frac{JB}{ne} = -RHJB$$

